

CARLA C. TY

📍 Intramuros, Metro Manila, Philippines ✉ carlaty07@gmail.com ☎ 09219358355 🌐 in/carla-4569911a3 🐙 tycarla.github.io

SUMMARY

Innovative and detail-oriented Computer Engineering undergraduate with hands-on experience in embedded systems, embedded software, IoT development, and circuit design. Experience in integrating hardware and software to build efficient, real-world solutions. Gained strong communication, problem-solving, adaptability, and conflict resolution skills through customer service roles, enhancing the ability to work effectively in fast-paced, collaborative environments. Passionate about designing scalable systems and eager to contribute to impactful technology projects while continuing to grow in the tech industry.

EXPERIENCE

CUSTOMER SERVICE REPRESENTATIVE EXPERIENCE (TELECOMMUNICATION ACCOUNT)

Alorica

May 2023 - SEPTEMBER 2023, Cubao, Quezon City

- Resolved 30+ customer inquiries daily with professionalism and speed
- Communicated technical details clearly to non-technical users
- Demonstrated reliability by maintaining a 95% schedule adherence rate
- Collaborated with other departments to improve customer support workflows

PROJECT

IMAGE PROCESSING BAMBOO USING YOLO ALGORITHM

IEEEExplore • May 2024 - May 2025

- Develop a device to distinguish between running and clumping bamboo growth habits using image data
- Utilize the YOLOv5 algorithm to automatically detect and classify bamboo growth habits patterns, allowing for efficient processing of image-based datasets
- Analyzed the performance of the created model using a confusion matrix
- Evaluate the practical applicability of the device and algorithm in real-world bamboo habitats.

VENDING MACHINE USING ATMEGA328P

ICpEP - MAPUA • May 2025 - May 2025

- Developed a coin-operated printing vending machine using an ATmega328P microcontroller
- Implement a combinational circuit for dispensing change, returning 2 pesos when a 5-peso coin is inserted.
- Accepts 1-peso and 5-peso coins with a coin slot module, dispensing forms priced at 3 pesos each.
- Features a user interface with a keyboard for entering required information on the form before printing.
- Uses a 7-segment display for a synchronous countdown of pages printed, controlled via a JK flip-flop-based 4-bit down counter.
- Equipped with a sensor and buzzer to alert when paper supply is empty.

AWSCC-MAPUA OFFICAL WEBSITE (BACKEND DEVELOPER)

mapua.awscceph.org • November 2024 - May 2025

- A collaboration work with other members of AWSCC- MAPUA to make a website for AWS Club Mapua.
- Designed and implemented a serverless web feature using **AWS DynamoDB, S3, and Lambda** to manage officer profiles.
- Developed an **HTML-based Lambda form** for uploading officer photos to **S3** and submitting profile data (full name, position, description, LinkedIn link) to **DynamoDB**.
- Integrated **CRUD operations** via Lambda functions to manage profile data.
- Built and tested the entire system **locally using AWS SAM**, with code organized in a self-contained repository prepared for forking and deployment.
- AWS Service: API Gateway, Lambda, S3, DynamoDB, IAM

MINI SMART HOME SECURITY SYSTEM

Mapua University

- Buzzer: Alerts occupants to intrusions.
- Temperature sensor: Detects fire or extreme temperatures.
- LEDs: Light up when the main door opens.
- Stepper/servo motors: Automate curtains and door control.
- Keypad: Allows secure access with password entry.
- Ultrasonic sensor: Detects movement or obstacles, activating the buzzer.
- Integration: These components create a smart, secure, and automated home system.

IOT WITH MINI SMART MEDICINE CABINET

Mapua University

- Integrates sensor technology for effective medication management and storage.
- Temperature sensor ensures optimal storage conditions, maintaining medication potency.
- Ultrasonic sensor tracks cabinet access by logging door openings and closings for usage monitoring and security.
- IR sensor monitors the quantity of medicines, helping users keep track of stock levels.
- Automated reminders through a buzzer and LED indicators alert users to low stock or if the cabinet is left open.
- Prioritizes medication adherence and safety, preventing errors and maintaining proper storage conditions.

MINI SMART BICYCLE FOR CYCLIST SAFETY

Mapua University

- Addresses the need for cyclist protection using ultrasonic sensors, LED indicators, a microcontroller Arduino Uno, LCD display, and a buzzer.
- Alerts cyclists about approaching vehicles from behind with a multi-faceted warning system.
- Incorporates a manual turning signal system for enhanced situational awareness.
- Features three buttons for signaling: left, right, and off.
 - - Pressing left or right triggers corresponding blinking yellow LEDs to indicate direction.
 - - The middle button turns off the signal.
- LCD display shows safety recommendations and messages to the cyclist.

EDUCATION

BACHELOR OF SCIENCE IN COMPUTER ENGINEERING

Minor in EMBEDDED SYSTEM TECHNOLOGY (ONGOING) • Mapua University • Intramuros Manila • AUGUST 2020 – DECEMBER 2025

CERTIFICATIONS

OPERATING SYSTEMS AND YOU: BECOMING A POWER USER COURSERA ISSUED

COURSERA • APRIL 2022

SYSTEM ADMINISTRATION AND IT INFRASTRUCTURE SERVICES COURSERA ISSUED

COURSERA • OCTOBER 2022

TECHNICAL SUPPORT FUNDAMENTALS GOOGLE ISSUED

GOOGLE • MAR 2021

THE BITS AND BYTES OF COMPUTER NETWORKING COURSERA ISSUED

COURSERA • MAY 2021

INTRODUCTION TO GIT AND GITHUB COURSERA ISSUED

COURSERA • MAY 2021

APPLIED PLOTTING, CHARTING & DATA REPRESENTATION IN PYTHON COURSERA ISSUED

COURSERA • OCTOBER 2022

SKILLS

Arduino UNO, ESP32, ESP8266, Raspberry Pi 4, ATMEGA328P

C/C++, Python, JavaScript

PCB Design, Circuit Schematic Design, Sensor Integration

Git/GitHub, VS Code, Anaconda, Jupyter Notebook, Docker

AWS IoT, HTTP/HTTPS APIs, RESTful Services, DynamoDB, AWS Lambda, AWS IAM, AWS S3

Embedded C, RTOS Basics

Proteus, Tinkercad

I2C, Bluetooth, WiFi

Arduino IDE
